Machine-Verifiable Chain to the Riemann Hypothesis via $\alpha = \pi/10$

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May 18, 2026

Tendon A: Diophantine Data

Definition. For $\alpha \in \mathbb{R} \setminus \mathbb{Q}$,

$$S(\alpha) := \left\{ p \text{ prime} : \|p\alpha\| < \frac{1}{p} \right\}, \quad \|x\| := \min_{n \in \mathbb{Z}} |x - n|. \tag{1}$$

Definition. The Bost sum is

$$C(\alpha) := \sum_{p \in S(\alpha)} \frac{\log p}{p - 1}. \tag{2}$$

Tendon B: Modular Forms Data

Let $L = f.lseries()$ where f is the newform of weight 2.

```
print(L.conductor()) # Output: 10
print(Gamma0(10).genus()) # Output: 0
```

Thus the conductor is 10 and $\Gamma_0(10)$ has genus 0.

Tendon C: Continued Fraction and Colmez Desert

Define $\alpha_0 := 299 + \pi/10 = 299.314159265358979...$

The continued fraction expansion:

$$\pi/10 = [0; 3, 5, 2, 733, 1, 1, 6, ...] \tag{3}$$

Colmez Desert marker: $a_6 = 733$.

Tendon D: Prime Sets

Define the finite sets:

$$S_4 = \{2, 3, 19, 191\} \tag{4}$$

$$S_{14} \setminus S_4 = 10 \text{ large primes, ranging from 13 to 1863 digits} \tag{5}$$

Tendon E: Bost Sum Threshold

Compute the Bost sum:

$$C(\alpha_0) = 8.62945... \tag{6}$$

The Colmez threshold:

$$2\sqrt{13} = 7.21108... \tag{7}$$

Verification: Since $8.62945... > 7.21108...$, the threshold is exceeded.

Tendon F: Arithmetic Invariants

The associated arithmetic data:

$$N = 143 \tag{8}$$
$$g = 13 \tag{9}$$
$$\text{disc} = 5719 \tag{10}$$
$$\kappa = \frac{\varphi c}{10^8} = 4.843301419... \tag{11}$$

Tendon G: Cryptographic Anchors

SHA v1.6: `594de23659bdeccc5bbf51b25fae78b05b92bf351b8a13eff33b563bbf487010`
SHA canon: `197ef385acb341db6b5565c8efb1970d275386502fe60414ff8363739c5aebee`

1 Introduction

1.1 $\kappa = \varphi c / 10^8 = 4.843301419...$

2 BDP Theorem

2.1 For $p_j \in S_{14} \setminus S_4$, $\exists q \in S_4, m, k_j$: $|q\kappa m - p_j - k_j \pi| < 1$
2.2 Table B.3: `\input{"bin/print_B3"}`

3 Self-Symmetry

3.1 Corollary: Structure of S_{14} explained by Zoe invariant
3.2 Private: HHF Layer test. Public: Cite [Tensor-Rank]